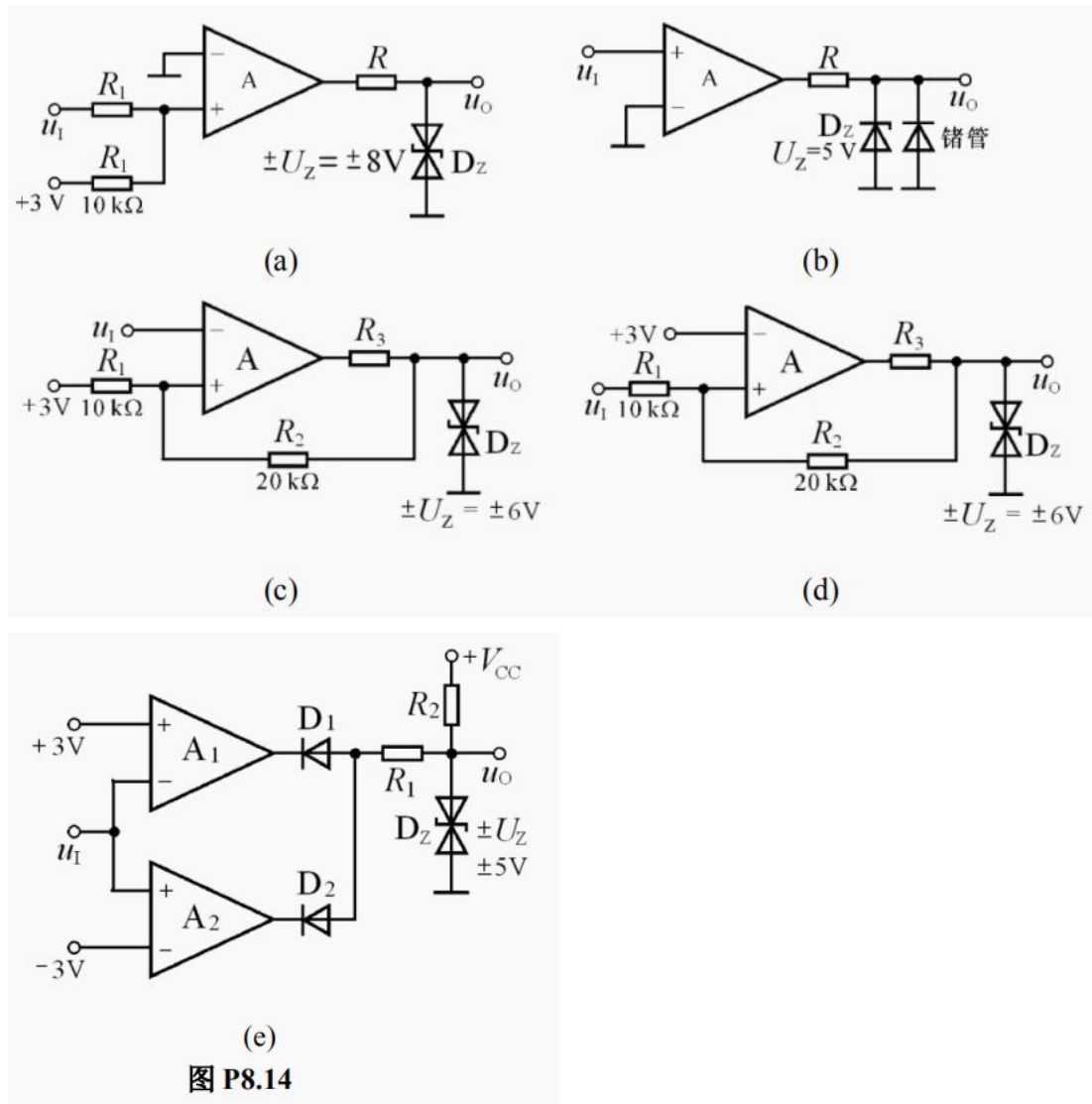


Homework 10

✧ Voltage comparator P473: 8.14/8.15/8.16/ 8.18/8.X

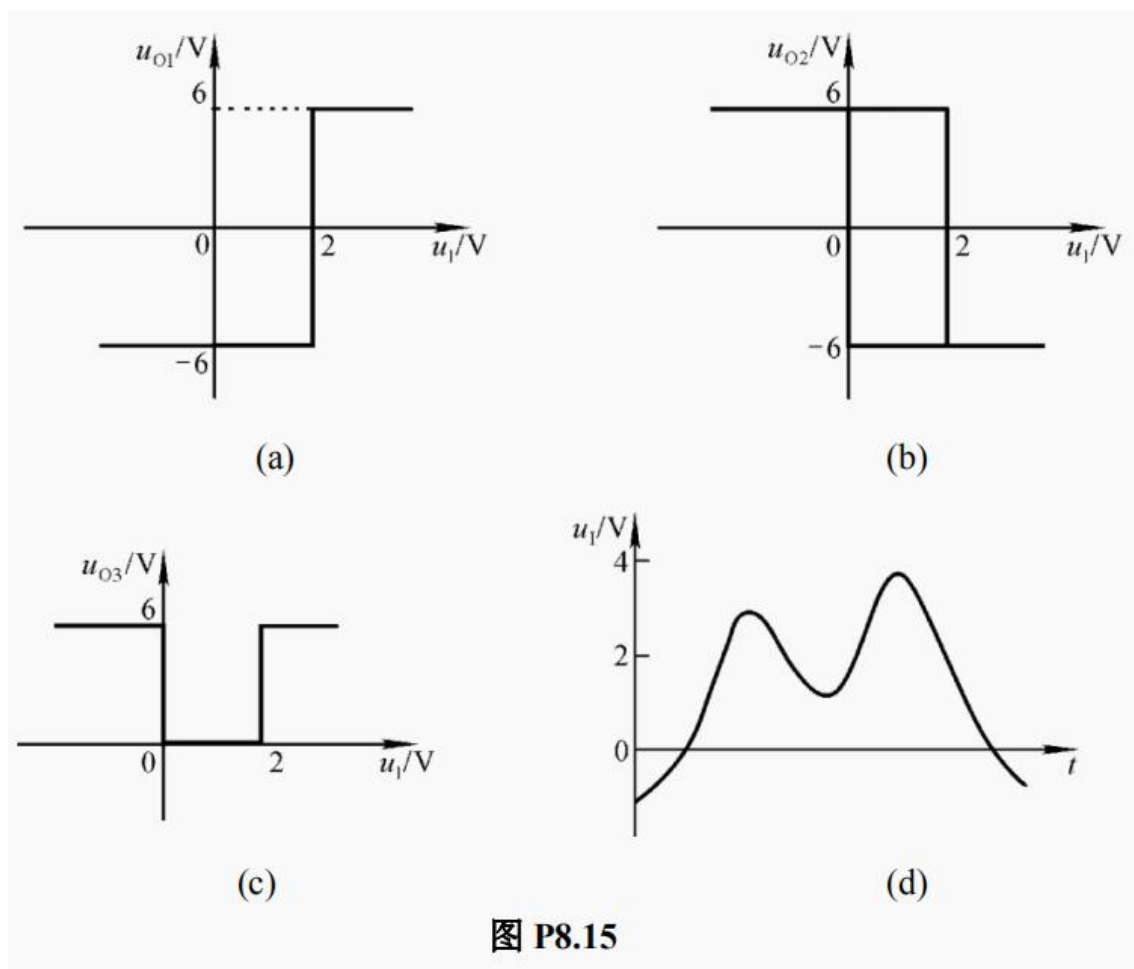
8.14 试分别求解图 P8.14 所示各电路的电压传输特性。

8.14 Please solve the voltage transmission characteristics ($u_o \sim u_i$) of each circuit as shown in figure p8.14.



8.15 已知三个电压比较器的电压传输特性分别如图 P8.15(a)、(b)、(c)所示，它们的输入电压波形均如图(d)所示，试画出 u_{o1} 、 u_{o2} 和 u_{o3} 的波形。

8.15 The voltage transmission characteristics of the three voltage comparators are shown in figure p8.15 (a), (b) and (c), respectively. The waveform of input voltage is shown in figure (d). Please draw the waveforms of output voltages u_{o1} 、 u_{o2} and u_{o3} .



8.16 图 P8.16 所示为光控电路的一部分，它将连续变化的光电信号转换成离散信号（即不是高电平，就是低电平），电流 i_l 随光照的强弱而变化。

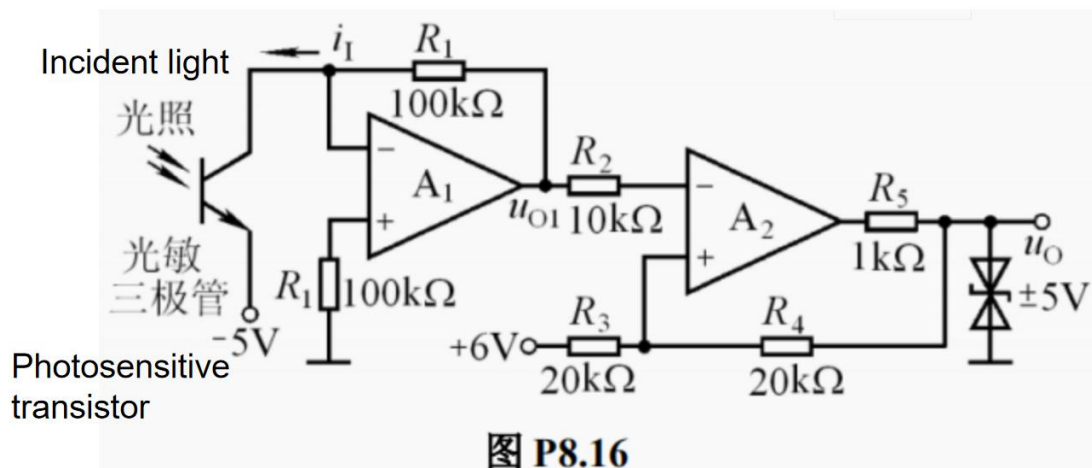
(1) 在 A_1 和 A_2 中，哪个工作在线性区？哪个工作在非线性区？为什么？

(2) 试求出表示 u_o 与 i_l 关系的传输特性。

8.16 Figure p8.16 is part of an optical control circuit, which converts the continuous photoelectric signal into discrete signal (i.e. high level or low level), and the current i_l changes with the intensity of the incident light. Ask:

(1) For operational amplifiers A_1 and A_2 , which one works in the linear amplification region and which works in the nonlinear saturation region? Why?

(2) Please draw the transmission characteristics of this circuit (u_o as function of i_l).

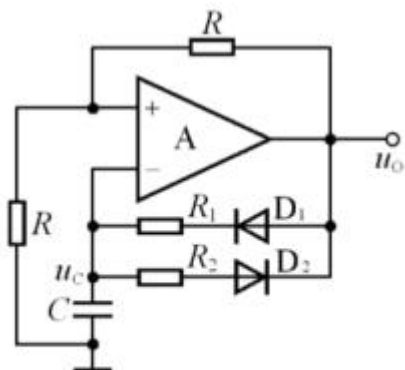


8.18 在图 P8.18 所示电路中, 已知 $R_1=10k\Omega$, $R_2=20k\Omega$, $C = 0.01\mu F$, 集成运放的最大输出电压幅值为 $\pm 12V$, 二极管的动态电阻可忽略不计。

- (1) 求出电路的振荡周期;
- (2) 画出 u_o 和 u_c 的波形。

8.18 In the circuit shown in figure P8.18, $R_1=10k\Omega$, $R_2=20k\Omega$ and $C = 0.01\mu F$. The maximum output voltage amplitude of the integrated op amp is $\pm 12V$, the dynamic resistance of diode can be ignored.

- (1) Calculate the oscillation period of the circuit;
- (2) Draw the waveform of u_o and u_c .



8. X 电路如图所示, 试分析:

- (1) 分别说明 A_1 和 A_2 各构成哪种基本电路;
- (2) 求出 u_{O1} 与 u_o 的关系曲线 $u_{O1}=f(u_o)$;
- (3) 求出 u_o 与 u_{O1} 的运算关系式 $u_o=f(u_{O1})$;
- (4) 定性画出 u_{O1} 与 u_o 的波形;
- (5) 说明若要提高振荡频率, 则可以改变哪些电路参数, 如何改变?

8. X As shown in the figure below.

- (1) Which types of fundamental circuits are A_1 and A_2 ;
- (2) Find the relationship $u_{O1}=f(u_o)$;
- (3) Find the calculation relation $u_o=f(u_{O1})$;

- (4) Draw the waveforms of u_{o1} and u_o ;
- (5) Explain which circuit parameters can be changed and how to change them if the oscillation frequency is to be increased?

